POINT OF SALE SYSTEM

Database Design Document
V 1.0

By

01	*Group Member 1 Muhammad Shoaib*	*Num-Bscs-2024-57*
02	*Group Member 2 Allah Ditta*	*Num-Bscs-2024-10*
03	*Group Member 3 Shahbaz Ali*	*Num-Bscs-2024-73*

Department of Computer Sciences
Namal University
Mianwali, Pakistan

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Contents

1.1. INTRODUCTION

This document presents the database design for a Point of Sale (POS) System for a pizza shop with multiple branches. The system helps manage daily operations such as orders, inventory, staff, customer transactions, deals, and menu items

. The system supports the following main processes:

- Taking customer orders through receptionists
- Assigning orders to kitchen staff based on current workload
- Tracking inventory in different branches
- Managing staff attendance, shifts, and salaries
- Applying discounts
- Maintaining menu item prices
- Tracking order status from pending to completed
- Managing branch wise deals and branch wise menu items

The system operates in a real-time environment where multiple users, including Admins, Receptionists, and Kitchen Staff, can use the system simultaneously across different branches. Each branch manages its own inventory stock and menu availability while sharing a common menu and pricing structure across all branches.

1.2. PROBLEM STATEMENT

Many pizza shops still use manual or semi-automated systems, which create several operational problems and reduce customer satisfaction.

- **Lack of Real-time Inventory Tracking:** Staff cannot accurately track current stock levels, which may result in orders being placed for unavailable items.
- **Inefficient Order Assignment:** Orders are assigned without considering the workload of kitchen staff, causing delays and uneven distribution of work.
- **No Price History:** When menu prices are updated, previous orders may show incorrect prices instead of the actual amount paid by customers.
- **Poor Accountability:** The system cannot identify whether mistakes in an order were made by the receptionist or the kitchen staff.
- **Limited Order Modification:** Customers cannot modify or cancel orders before preparation starts.

- **No Multi-branch Support:** Branches operate separately without a centralized system for managing inventory, staff, and orders.
- **Per-Branch Menu Control:** Menu items cannot be marked unavailable for one branch without affecting other branches.

These issues can lead to wasted inventory, delays in service, reduced customer satisfaction, revenue loss, and inefficient staff management.

1.3. PROJECT OBJECTIVES

General Objective: To design and implement an efficient and user friendly Point of Sale System that automates order processing, tracks inventory in real time provides accountability for transactions, and supports multiple branches.

Specific Objectives:

- **Automate Order Assignment:** Automatically assign new orders to the kitchen staff member with the fewest active orders to balance workload.
- **Enforce Inventory Validation:** Prevent orders when the requested quantity exceeds available stock, while supporting partial order options.
- **Maintain Complete Price History:** Store historical prices of menu items to keep past order records accurate.
- **Implement Dual Accountability:** Track both the receptionist who takes the order and the kitchen staff who prepares it.
- **Support Order Modification:** Allow customers to modify or cancel orders before preparation begins.
- **Enable Multi-branch Operation:** Support multiple branches with separate inventory tracking, menu and deal availability.
- **Generate Low Stock Alerts:** Notify staff automatically when inventory falls below reorder levels.
- **Track Staff Attendance:** Record clock in and clock out times of staff members.
- **Manage Staff Salaries:** Maintain salary history with admin-only access.
- **Support Deals:** Allow creation and management of deals with branch wise availability.
- **Per Branch Menu Control:** Allow each branch to manage its own menu availability independently.

1.4. GITHUB REPOSITORY LINK

https://github.com/muhammadshoaibnoor/Database-System-Project.git

2.1. ENTITY

The following entities have been identified as core to the POS and Inventory Management System:

Sr. No	Entity Name	Description
01	Branch	Represents a physical store location with its own address, contact information, and operating hours.
02	User	Represents staff members including Admin, Receptionist, and Kitchen Staff who interact with the system.
03	Salary	Records payments made to staff members including amount, date, and payment method.
04	Shift	Tracks staff attendance including start time and end times.
05	Category	Groups menu items into logical categories such as Pizza, Drinks, or Desserts.
06	Menu_Item	Represents products sold including pizzas, add-ons, and cold drinks.
07	Price	Maintains historical pricing for each menu item with effective dates.
08	Order	Represents a customer transaction including order status, payment status, and cash handling.
09	Order_Item	Represents individual line items within an order including quantity and unit price.
10	Inventory	Tracks raw materials stock levels per branch including reorder thresholds.
11	Menu_Ingredient	Defines the ingredients and quantities required to prepare each menu item.
12	Discount	Defines discount with discount type, value, and validity period.
13	Deal	Represents combo offers that bundle multiple menu items together.
14	Deal_Item	Links deals to the menu items they contain with quantities.
15	Branch_Menu_Item	Manages per branch menu item availability.
16	Branch_Deal	Manages per branch deal availability.

2.2. DATA DICTIONARY

2.2.1. Branch

Sr. No	Name	Data Type	Constraint	Description
01	BranchId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each branch
02	BranchName	VARCHAR(100)	NOT NULL, UNIQUE	Name of the branch location
03	Address	TEXT	NOT NULL	Physical address of the branch
04	PhoneNumber	VARCHAR(20)	NOT NULL	Contact phone number
05	Email	VARCHAR(100)	UNIQUE	Email address of the branch
06	OpeningTime	TIME	NOT NULL	Daily opening time
07	ClosingTime	TIME	NOT NULL	Daily closing time
08	IsActive	BOOLEAN	DEFAULT TRUE	Whether branch is operational

2.2.2. User

Sr. No	Name	Data Type	Constraint	Description
01	UserId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each user
02	FirstName	VARCHAR(50)	NOT NULL	First name of the staff member
03	LastName	VARCHAR(50)	NOT NULL	Last name of the staff member
04	Email	VARCHAR(100)	NOT NULL, UNIQUE	Email address
05	PhoneNumber	VARCHAR(20)	NOT NULL	Contact phone number
06	Username	VARCHAR(50)	NOT NULL, UNIQUE	Login username
07	Password	VARCHAR(255)	NOT NULL	Hashed password
08	Role	VARCHAR(20)	NOT NULL	Role: Admin, Receptionist, Kitchen Staff
09	IsActive	BOOLEAN	DEFAULT TRUE	Whether user is currently employed
10	HireDate	DATE	NOT NULL	Date when staff was hired
11	ResignationDate	DATE	NULL	Date when staff left (NULL if active)

2.2.3. Salary

Sr. No	Name	Data Type	Constraint	Description
01	SalaryId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each salary payment
02	UserId	INT	NOT NULL	References the user who received payment
03	Amount	DECIMAL(10,2)	NOT NULL	Salary amount paid
04	PaymentDate	DATE	NOT NULL	Date when payment was made
05	PaymentMethod	VARCHAR(20)	NULL	Method: Cash, Bank Transfer, Check
06	Notes	TEXT	NULL	Additional notes about the payment

2.2.4. Shift

Sr. No	Name	Data Type	Constraint	Description
01	ShiftId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each shift
02	UserId	INT	NOT NULL	References the user who worked the shift
03	ClockIn	DATETIME	NOT NULL	Date and time when staff clocked in
04	ClockOut	DATETIME	NULL	Date and time when staff clocked out

2.2.5. Category

Sr. No	Name	Data Type	Constraint	Description
01	CategoryId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each category
02	CategoryName	VARCHAR(50)	NOT NULL, UNIQUE	Name of the category (e.g., Pizza, Drinks)

2.2.6. Menu_Item

Sr. No	Name	Data Type	Constraint	Description
01	MenuItemId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each menu item
02	ItemType	VARCHAR(20)	NOT NULL	Type: Pizza, AddOn, Coldrink
03	ItemName	VARCHAR(100)	NOT NULL	Name of the menu item
04	Description	TEXT	NULL	Detailed description of the item
05	CategoryId	INT	NOT NULL	References the category this item belongs to

2.2.7. Price

Sr. No	Name	Data Type	Constraint	Description
01	PriceId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each price record
02	MenuItemId	INT	NOT NULL	References the menu item
03	Amount	DECIMAL(10,2)	NOT NULL	Price amount
04	EffectiveDate	DATETIME	NOT NULL	Date when this price becomes active
05	EndDate	DATETIME	NULL	Date when this price expires (NULL = current)
06	IsActive	BOOLEAN	DEFAULT FALSE	Whether this is the active price

2.2.8. Branch_Menu_Item

Sr. No	Name	Data Type	Constraint	Description
01	BranchId	INT	PRIMARY KEY, NOT NULL	References the branch
02	MenuItemId	INT	PRIMARY KEY, NOT NULL	References the menu item
03	IsAvailable	BOOLEAN	DEFAULT TRUE	Whether this menu item is available at this branch

2.2.9. Deal

Sr. No	Name	Data Type	Constraint	Description
01	DealId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each deal
02	DealName	VARCHAR(100)	NOT NULL	Name of the deal/combo
03	TotalAmount	DECIMAL(10,2)	NOT NULL	Total price of the deal
04	IsActive	BOOLEAN	DEFAULT TRUE	Whether the deal is currently active

2.2.10. Deal_Item

Sr. No	Name	Data Type	Constraint	Description
01	DealId	INT	PRIMARY KEY, NOT NULL	References the deal
02	MenuItemId	INT	PRIMARY KEY, NOT NULL	References the menu item in the deal
03	Quantity	INT	NOT NULL, DEFAULT 1	Quantity of this menu item in the deal

2.2.11. Branch_Deal

Sr. No	Name	Data Type	Constraint	Description
01	BranchId	INT	PRIMARY KEY, NOT NULL	References the branch
02	DealId	INT	PRIMARY KEY, NOT NULL	References the deal
03	IsAvailable	BOOLEAN	DEFAULT TRUE	Whether this deal is available at this branch

2.2.12. Order

Sr. No	Name	Data Type	Constraint	Description
01	OrderId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each order
02	UserId	INT	NOT NULL	Receptionist who took the order
03	PreparedBy	INT	NULL	Kitchen staff who prepared the order
04	DiscountId	INT	NULL	Discount applied to this order
05	DiscountAmount	DECIMAL(10,2)	NULL	Amount of discount applied
06	TotalAmount	DECIMAL(10,2)	NOT NULL	Final total after discount
07	OrderDate	DATETIME	NOT NULL	Date and time when order was placed
08	Status	VARCHAR(20)	NOT NULL	Status: Pending, In Preparation, Completed, Cancelled
09	PaymentStatus	VARCHAR(20)	NOT NULL	Status: Unpaid, Paid, Refunded
10	CashTendered	DECIMAL(10,2)	NULL	Amount of cash given by customer
11	ChangeDue	DECIMAL(10,2)	NULL	Change to return to customer

2.2.13. Order_Item

Sr. No	Name	Data Type	Constraint	Description
01	OrderItemId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each order item
02	OrderId	INT	NOT NULL	References the order
03	MenuItemId	INT	NULL	References menu item (for single items)
04	DealId	INT	NULL	References deal (for combo items)
05	Quantity	INT	NOT NULL	Quantity ordered
06	UnitPrice	DECIMAL(10,2)	NOT NULL	Price per unit at time of order
07	Subtotal	DECIMAL(10,2)	NOT NULL	Quantity * UnitPrice

2.2.14. Inventory

Sr. No	Name	Data Type	Constraint	Description
01	InventoryId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each inventory item
02	BranchId	INT	NOT NULL	References the branch
03	InventoryName	VARCHAR(100)	NOT NULL	Name of the raw material
04	Quantity	DECIMAL(10,2)	NOT NULL, DEFAULT 0	Current stock quantity
05	Unit	VARCHAR(20)	NOT NULL	Unit of measurement (kg, grams, pieces)
06	ReorderLevel	DECIMAL(10,2)	NOT NULL	Threshold for low stock alert
07	UnitCost	DECIMAL(10,2)	NULL	Cost per unit

2.2.15. Menu_Ingredient

Sr. No	Name	Data Type	Constraint	Description
01	MenuIngredientId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each recipe record
02	MenuItemId	INT	NOT NULL	References the menu item
03	InventoryId	INT	NOT NULL	References the inventory item
04	QuantityRequired	DECIMAL(10,2)	NOT NULL	Amount of inventory required per unit

2.2.16. Discount

Sr. No	Name	Data Type	Constraint	Description
01	DiscountId	INT	PRIMARY KEY, NOT NULL	Unique identifier for each discount
02	DiscountName	VARCHAR(100)	NOT NULL	Name of the discount
03	DiscountType	VARCHAR(20)	NOT NULL	Type: Percentage, Fixed Amount
04	DiscountValue	DECIMAL(10,2)	NOT NULL	Value of discount (10 for 10%, or 5.00)
05	StartDate	DATETIME	NOT NULL	Date when discount becomes active
06	EndDate	DATETIME	NULL	Date when discount expires
07	IsActive	BOOLEAN	DEFAULT TRUE	Whether discount is currently active

2.3. BUSINESS RULES

The following business rules define how the POS and Inventory Management System operates:

2.3.1. Branch & Staff Rules

- A branch can have many users (staff members).
- A branch can exist without any users (new branch).
- One admin can manage many users.
- Each user belongs to exactly one branch.
- Users can be Admin, Receptionist, or Kitchen Staff.
- Different branches can have different menu items based on area customer preferences.
- A menu item may be unavailable at a specific branch due to inventory shortage while being available at other branches.
- Only Admin can view the Salary table. Receptionists and Kitchen Staff cannot access salary information.

2.3.2. Staff & Shift Rules

- One user can work many shifts over time.
- A user may have no shifts (new hire).
- Each shift belongs to exactly one user.
- Shifts track clock in and clock out times.

2.3.3. Salary Rules

- One user has many salary payment records.
- Each salary payment belongs to exactly one user.
- Only Admin can view salary information.

2.3.4. Menu & Category Rules

- One category contains many menu items.
- Each menu item belongs to exactly one category.
- Menu items can be Pizza, AddOn, or Coldrink.
- One menu item has many price records (price history).
- Each price record belongs to exactly one menu item.
- Only one active price per menu item at any given time.

2.3.5. Order Rules

- One receptionist can take many orders.
- One kitchen staff can prepare many orders.
- Each order is taken by exactly one receptionist.
- Each order is prepared by exactly one kitchen staff.
- One order contains many order items.
- Each order item belongs to exactly one order.
- When an order is in PENDING status, the customer can modify or cancel the order.
- When an order is modified, existing order items are overwritten.
- When an order is cancelled in PENDING status, a full refund is issued.
- When an order is in IN PREPARATION status, it cannot be modified or cancelled.
- If a receptionist takes a wrong order, the receptionist is accountable.
- If kitchen staff burns or prepares an order incorrectly, the kitchen staff is accountable.

2.3.6. Order Assignment Rules

- Orders are assigned to the kitchen staff member with the fewest active orders.
- If multiple staff have same active orders, assign to lowest UserId (first PC).

2.3.7. Discount Rules

- One discount can apply to many orders.
- Each order can have at most one discount (zero or one).
- Discounts have a start date and end date.
- Only active discounts within valid date range can be applied.

2.3.8. Inventory Rules

- One branch has many inventory items.
- A branch can have zero inventory items (new branch).
- An order cannot be placed if requested quantity exceeds available stock.
- Low stock notifications are sent to all staff (real-time, not stored).
- An add-on cannot be added if its inventory is low.

2.3.9. Deal Rules

- A deal can contain many menu items.
- One menu item can be included in many deals.
- When a deal is ordered, all its items subtract from inventory.
- Customer can order deals and single items in same order.
- Customer can request add-ons when ordering a deal.
- If any menu item in a deal is unavailable at a branch, the entire deal becomes unavailable at that branch.

2.4. ENTITY RELATIONSHIP DIAGRAM

Figure 2.1 presents the complete Entity Relationship Diagram for the POS System. The diagram uses Crow's Foot notation to show all entities, attributes, primary keys, foreign keys, and cardinalities between entities.

The ERD contains 16 entities:

- Branch
- User
- Salary

- Shift
- Category
- Menu_Item
- Price
- Order
- Order_Item
- Inventory
- Recipe
- Discount
- Deal
- Deal_Item
- Branch_Menu_Item
- Branch_Deal

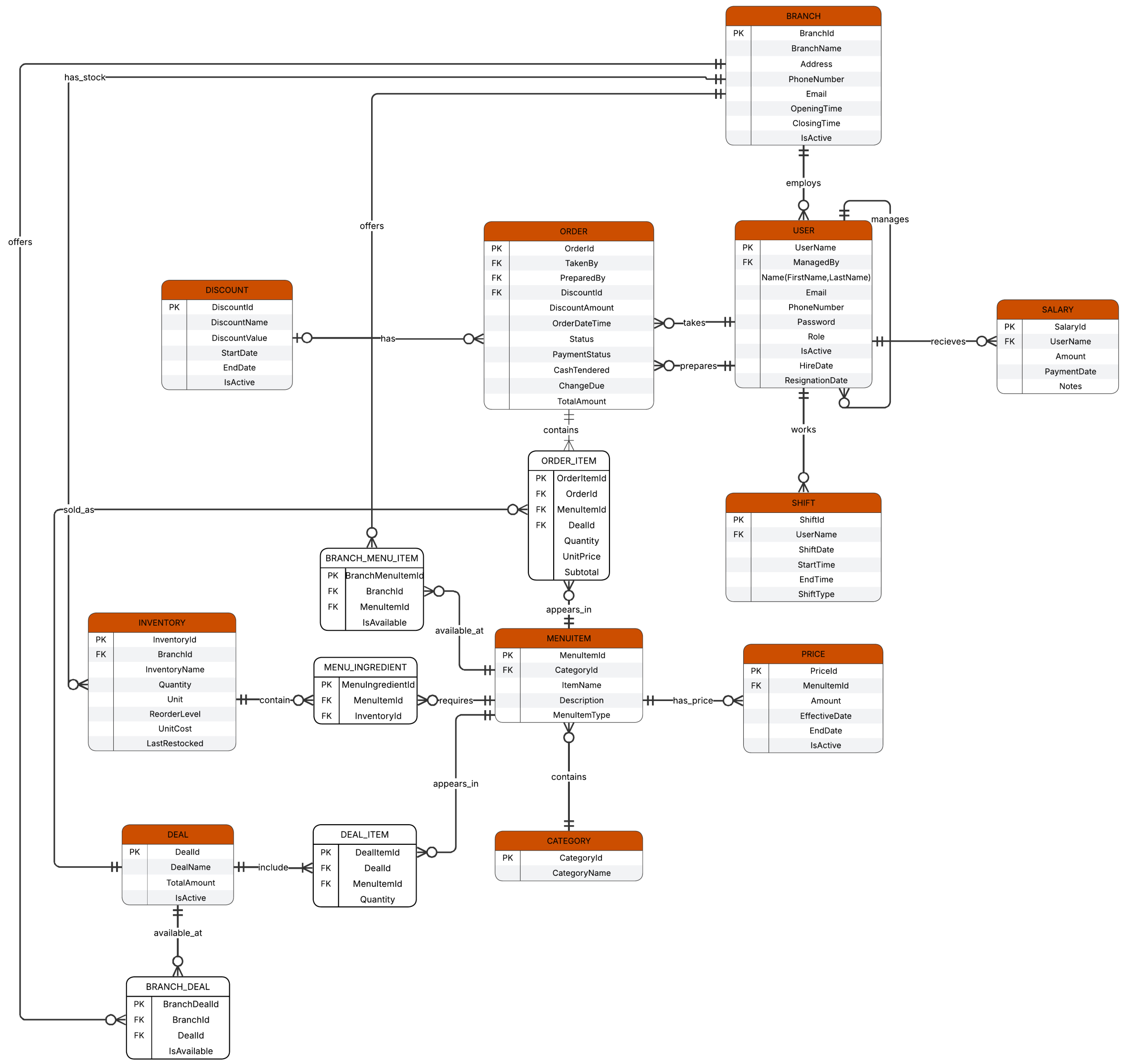

Figure 2.1: Entity Relationship Diagram of POS System

[1] "Lucidchart," Lucid Software Inc. https://www.lucidchart.com, 2024.

[2] "Overleaf," Overleaf. https://www.overleaf.com, 2024.

This section documents the AI tools and prompts utilized during the development of this project.

AI TOOL USED

- **Tool:** DeepSeek and Gemini
- **Purpose:** Learning assistance, design exploration, concept validation

IMPORTANT PROMPTS FOR SYSTEM DESIGN, ERD DEVELOPMENT & DOCUMENTAION

Sr. No	Prompt	Purpose
01	"Understand my database project. This ERD represents a POS and Inventory Management System using Chen's Notation."	Initial project understanding
02	"I want to add price entity."	Learning price history concept
03	"I want Add On entity for extra cheese and toppings."	Learning add-on design pattern
04	"Is when pizza has some addon, quantity deduct from inventory?"	Understanding inventory deduction for add-ons
05	"Does addon has relation with price entity?"	Learning price history for add-ons
06	"Add entity of shift for user."	Learning staff attendance tracking
07	"Add Branch entity. Each branch has 1 admin and 1 admin manages many users."	Learning multi-branch architecture
08	"I want salary entity."	Learning staff payment tracking
09	"Does we have to make a supertype and subtype?"	Understanding inheritance in databases
10	"Give me a plant uml code of complete erd in crow foot notation."	Learning ERD visualization
11	"Does we have to add discount entity?"	Learning discount design patterns
12	"Explain the relation, cardinality optional and mandatory between Deal, DealItem and MenuItem."	Learning relationship cardinalities
13	"Explain between Branch, BranchDeal and Deal - optional vs mandatory."	Understanding junction tables

14	"Why relation between branch and inventory is optional many?"	Understanding optional cardinalities
15	"Why branch has optional many user?"	Learning optional relationships
16	"Why menu item has no IsAvailable?"	Learning per-branch availability
17	"What business rules you know about my POS system?"	Business rule documentation
18	"When receptionist takes order, order assigned to kitchen staff with least orders. If same, go to first PC."	Learning order assignment logic
19	"Can we do this without changing database with C++ code?"	Learning application-level logic
20	"When order is pending, customer can modify or cancel. When In Preparation, cannot modify or cancel."	Learning order status workflow
21	"If receptionist takes wrong order, receptionist is blameable. If kitchen staff burns order, kitchen staff is blameable."	Learning accountability design
22	"When inventory is less than reorder level, notification sent to receptionist, kitchen staff, admin."	Understanding inventory alerts
23	"Order cannot be placed if inventory is low. If 4 pizzas available, order of 5 cannot be taken."	Learning inventory validation
24	"Low stock items not automatically removed. Receptionist removes manually if customer agrees to partial order."	Learning partial order handling
25	"Only one or zero discount applies on one order."	Understanding discount rules
26	"Order status can be only Pending, In Preparation, Completed, Cancelled."	Learning order status values
27	"Only admin can see salary table, not receptionist or kitchen staff."	Understanding access control
28	"Different branches have different menus based on area people choices."	Learning per-branch menu customization
29	"Fix ERD according to Multi-Branch Menu Control, Dynamic Pricing, Deals Entity."	Learning ERD corrections
30	"Is this ERD confirm that if a branch has low inventory for a menu item in a deal, the whole deal is unavailable?"	Learning deal availability logic
31	"Tell me best relationship names between entities."	Understanding relationship naming
32	"Why total amount is not in order? But I want to store it."	Learning derived attributes

33	”Explain the entities related to addon and why it is necessary?”	Understanding add-on architecture
34	”Is addoningredient associative entity?”	Learning associative entities
35	”How one addon use many inventory items?”	Understanding recipe concept
36	”One menu item has how many recipe? One recipe has how much inventory?”	Learning cardinalities
37	”Why we need category entity?”	Understanding categorization
38	”Explain salary entity. Does it tell old salaries?”	Learning salary history
39	”What is the purpose of shift entity?”	Understanding shift tracking
40	”Why applied at in OrderDiscount?”	Learning audit trail
41	”What is unit price at order in OrderItem?”	Learning price snapshot concept
42	”Why CreatedAt in branch and user? Add ResignationDate in user.”	Learning audit fields
43	”Customer can order deals + single items in same order. Can user request addon with deal?”	Learning combined orders
44	”When deal ordered with extra topping, does ingredients subtract from inventory?”	Understanding deal inventory deduction
45	”Is there subclass superclass can come? Is it efficient?”	Learning vs single table design
46	”Generate my detailed database document latex code.”	Knowing report structure and format

Prompts from Gemini

Sr. No	**Prompt**	**Purpose**
47	”Analyze this ERD and explain what you understand about the POS system.”	Initial project understanding, structural analysis, and core module identification
48	”Does you find any error in this ERD or a system?”	Vulnerability assessment, logic gap discovery, and database normalization check

49	"Explain this erd shortly in wording."	Extracting transactional flow and operational modules from Crow's foot notation image
50	"Write this Exactly this ERD Plant uml code."	Generating automated, styled database visualization code with specific cardinalities
51	"For 1 I will go with your suggestion, For 2 I am not satisfy with your suggestion so I keep it like previous, For 3 this is our business rule that we are not concern with this that payment is given by which payment method, For 4 I will go with your suggestion, Did not understand 5 and 6."	Setting business rules/constraints and requesting clarification on identity design and free item mechanics
52	"Not createdBy that is createdAt. and still I didnot understand 6."	Correcting image text interpretation (timestamp vs user ID) and exploring conditional sales logic
53	"No complementary item. But I am thinking that there should be a deal entity."	Designing a bundled sales architecture to track mixed promotions (standalone items vs combos)
54	"User can buy from single single menu item or he can buy from deal."	Designing a flexible, nullable foreign key shopping cart pattern (OrderItem) to map dual-purchasing paths
55	"Ok everything we discuss the errors and their fixes explain in one message."	Consolidating the final structural changes into a single development blueprint
56	"Remove complementary Item management."	Finalizing system constraints by removing unwanted features and polishing the production checklist
57	"In latex how to add on top chapter 1 and on each page different."	Implementing dynamic running headers utilizing the fancyhdr library

58	"Please give me latex code so document looks similar to the attached image."	Structuring template layout matching specific table borders, line rules, and text formats
59	"The chapter title and headings are printing twice in the body. Completely remove body chapter titles so it only shows up once in the top right header."	Debugging the underlying default LaTeX chapter macro engine and engineering a custom page-clearing bypass
60	"I want to remove 'Chapter 3' before References using unnumbered layout."	Creating a localized, reset macro wrapper to format header injections independently for appendix and index sections